06 – The decision-support system end to end

The five parts (prediction, target gap, explanation, recommendations, chatbot safety and university lookup) run for the three made-up example students. The chatbot's language model is not called here (it needs Ollama); its fixed safety and university answers are shown.

In [1]:
```
# Setup: work from the project root so the project code and configs are found
import os, sys, json, platform, warnings
from pathlib import Path
ROOT = Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd()
os.chdir(ROOT); sys.path.insert(0, str(ROOT))
warnings.filterwarnings("ignore")
import numpy as np, pandas as pd
from IPython.display import Image, Markdown, display
pd.set_option("display.precision", 3); pd.set_option("display.width", 160);
pd.set_option("display.max_columns", 20)
from src.utils.config import load_config, resolve_path
REPORTS = resolve_path(load_config("project")["paths"]["reports"])
FIG = resolve_path(load_config("project")["paths"]["figures"])
print("Project folder:", ROOT)
print("Python", platform.python_version(), "| numpy", np.__version__, "|
pandas", pd.__version__)
```

```
Project folder: C:\AL_outcome_predictor
Python 3.12.0 | numpy 2.5.4 | pandas 3.0.6
```

In [2]:
```
import yaml
from src.system import DecisionSupportSystem
examples =
yaml.safe_load(resolve_path("configs/example_profiles.yaml").read_text("utf-8"))
system = DecisionSupportSystem.load()
rows = []
reports = {}
for name, profile in examples.items():
    r = system.report(profile); reports[name] = r
    p, g, plan = r["prediction"], r["gap"], r["recommendations"]
    rows.append({"student": name, "target": profile["al_target_degree"],
"district": profile["al_district"],
                 "Z best guess": p["zscore"]["predicted"], "Z range": f"
{p['zscore']['range_low']}–{p['zscore']['range_high']}",
                 "band": p["zscore_band"]["most_likely"], "top degree area":
p["degree_category"]["most_likely"],
                 "gap status": g["status"], "gap": g.get("gap_points"), "plan
mode": plan["mode"], "suggestions": len(plan["recommendations"])})
pd.DataFrame(rows).set_index("student")
```

Out[2]:

student	target	district	Z best guess	Z range	band	top degree area	gap status	gap	plan mode	suggestions
strong	Medicine	Kandy	2.33	2.02–2.67	Band 3	Medicine	on_track	0.00	strengthen	4
average	Engineering	Gampaha	1.42	1.11–1.75	Band 2	Physical Science	gap	0.11	close_the_gap	7
weak	Medicine	Colombo	1.12	0.81–1.46	Band 1	Biological Science	gap	0.76	close_the_gap	11

Target gap messages

In [3]:
```
for name, r in reports.items():
    print(f"{name}: {r['gap']['message']}")
```

```
strong: Your predicted Z-score (2.33) reaches the lowest 2023/2024 cut-off for Medicine in your district (Medicine, Wayamba University of Sri Lanka: 1.78). Keep going to make it safer.
average: Your predicted Z-score (1.42) is 0.11 below the lowest 2023/2024 cut-off for Engineering in your district (Engineering (EM), University of Moratuwa: 1.53). A study plan can help close this gap.
weak: Your predicted Z-score (1.12) is 0.76 below the lowest 2023/2024 cut-off for Medicine in your district (Medicine, University of Moratuwa: 1.88). A study plan can help close this gap.
```

First three study suggestions per student

In [4]:
```
for name, r in reports.items():
    print(name + ":", " | ".join(x["title"] for x in r["recommendations"]
["recommendations"][:3]))
```

```
strong: Set a weekly self-study target | Practise past papers | Revise earlier lessons regularly
average: Explore your career goal | Give every study session one clear goal | Do a weekly catch-up
weak: Explore your career goal | Strengthen your Science basics early | Build confidence step by step
```

Chatbot: safety guard and exact university lookup (no language model needed)

In [5]:
```
from src.chatbot.guard import topic_of
from src.chatbot.lookup import university_answer
for message in ("Which tablets help me study longer?", "Who won the cricket
match yesterday?", "What should I do first?"):
    print(f"{message!r:45} -> topic: {topic_of(message)}")
print()
print(university_answer("Am I close to University of Peradeniya?",
reports["average"]))
```

```
'Which tablets help me study longer?'         -> topic: medical
'Who won the cricket match yesterday?'        -> topic: unrelated
'What should I do first?'                     -> topic: in_scope
```

```
Engineering at University of Peradeniya: the 2023/2024 cut-off for Gampaha
district was 1.9259. Your predicted Z-score (1.42) is 0.51 below it, so it is
further away for now: your whole likely range is below last year's cut-off. This
is a guide from last year's cut-offs, not a promise: cut-offs change every year,
and the UGC also uses the order in which you list universities.
```